

Difficulty-Stratified Evaluation of Confidence Calibration in Large Language Models

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Abstract

Large language models (LLMs) are increasingly deployed as decision-support systems, yet the fluency of their responses can mask substantial uncertainty. Although calibration is essential for trustworthy AI, limited work has examined self-reported confidence across controlled task difficulty while comparing proprietary and open-weight models. We evaluate Claude Haiku, Llama 3.1, and Mistral on a frozen, difficulty-stratified subset of the Massive Multitask Language Understanding (MMLU) benchmark containing 180 multiple-choice questions across STEM, Social Sciences, and Humanities, with easy, medium, and hard difficulty levels. Each model completed two trials at a fixed temperature of 0.3 and provided both an answer and confidence estimate (0–100). Calibration was evaluated using accuracy, overconfidence gap (mean confidence minus accuracy), Expected Calibration Error (ECE), and difficulty-dependent overconfidence trends, excluding unparseable responses. All models exhibited increasing overconfidence as difficulty increased, consistent with the hard–easy effect observed in human judgment. Claude Haiku achieved strong calibration (92.1% accuracy; ECE = 0.05) with slight overall underconfidence, while Llama 3.1 (63.6% accuracy; ECE = 0.23) and Mistral (53.8% accuracy; ECE = 0.45) showed substantial overconfidence, particularly on hard questions. These results demonstrate that verbal confidence is not a reliable proxy for correctness, especially for weaker open-weight models on challenging tasks. Self-reported confidence should therefore complement, not replace, rigorous calibration methods and human oversight in high-stakes applications. Limitations include the use of multiple-choice evaluation, reliance on verbal rather than probability-based confidence estimates, and reduced Claude Haiku parse rates due to output truncation.

Keywords Large Language Models (LLMs), Confidence Calibration, Task Difficulty, MMLU, Expected Calibration Error (ECE), Hard–Easy Effect.

1. INTRODUCTION

Large language models (LLMs) are increasingly used as study aids, productivity assistants, and informal decision-support tools. Their responses are often fluent, coherent, and presented with an authoritative tone, which can encourage users to interpret them as reliable even when the underlying information is incorrect or uncertain. As these systems become more integrated into education, research, and professional workflows, understanding whether an LLM's expressed confidence accurately reflects its likelihood of being correct has become an important consideration for the development and deployment of trustworthy artificial intelligence.

Confidence calibration refers to the degree to which reported confidence aligns with observed accuracy. A well-calibrated system expresses high confidence when it is likely to be correct and lower confidence when uncertainty is warranted. In cognitive psychology, calibration has long been recognized as a fundamental aspect of human judgment, with confidence systematically varying according to task difficulty. People tend to overestimate their performance on difficult tasks while underestimating it on easier tasks, a phenomenon commonly referred to as the *hard–easy effect* (Moore & Healy, 2008). Kruger and Dunning (1999) further demonstrated that limited competence may be accompanied by limited insight into one's own performance,

leading individuals to express confidence that exceeds their actual ability. These findings suggest that confidence should be interpreted in the context of task difficulty rather than as an absolute indicator of correctness.

Confidence calibration is also an essential property of machine learning systems. Guo et al. (2017) demonstrated that modern neural networks are frequently miscalibrated and introduced Expected Calibration Error (ECE), which has become a standard metric for quantifying the agreement between confidence and accuracy. Unlike many traditional machine learning models, LLMs can express confidence directly through natural language, allowing calibration to be evaluated using verbally reported confidence rather than internal probability estimates (Kadavath et al., 2022; Xiong et al., 2024). As LLMs become increasingly relied upon for information retrieval, learning, and decision support, it is important to determine whether these verbal confidence estimates remain reliable across varying levels of task difficulty.

The purpose of this study is to examine how task difficulty influences verbal confidence calibration in large language models. Specifically, this study evaluates Claude Haiku, Llama 3.1, and Mistral using a frozen, difficulty-stratified subset of the Massive Multitask Language Understanding (MMLU) benchmark (Hendrycks et al., 2021). The evaluation consists of multiple-choice questions spanning STEM, Social Sciences, and Humanities, organized into pre-registered easy, medium, and hard difficulty tiers. Each model is required to provide both an answer and an explicit confidence estimate using a constrained response format. Calibration is assessed using accuracy, the overconfidence gap, and Expected Calibration Error to determine whether verbal confidence becomes increasingly misaligned with correctness as task difficulty increases and whether these patterns resemble the difficulty-dependent confidence effects observed in human judgment.

By evaluating calibration as a function of task difficulty rather than overall performance alone, this study provides a controlled assessment of verbal confidence across proprietary and open-weight language models under a common

experimental framework. Understanding how confidence changes as questions become more challenging has practical implications for the use of LLMs in educational, professional, and decision-support settings, where users may rely on expressed confidence as an indicator of reliability.

2. LITERATURE REVIEW

Calibration in Large Language Models

Confidence calibration has long been recognized as an essential property of reliable machine learning systems because predictive confidence should correspond closely to empirical accuracy. Early work in natural language processing demonstrated that confidence estimates produced by neural language models are not always well aligned with correctness. Desai and Durrett (2020) found that pretrained Transformer models exhibited substantially poorer calibration when evaluated on out-of-domain data than on in-domain tasks, indicating that model confidence deteriorates as inputs become less familiar. Similarly, Jiang et al. (2021) examined generative question answering models, including T5, BART, and GPT-2, and reported that these systems frequently produced overconfident predictions despite incorrect answers. Their work further showed that fine-tuning and post-hoc calibration methods could partially mitigate these discrepancies. Together, these studies established that strong predictive performance does not necessarily imply well-calibrated confidence, particularly under more challenging evaluation conditions.

Verbal Confidence and Uncertainty Estimation

Unlike traditional neural networks, large language models can communicate uncertainty directly through natural language. Lin et al. (2022) demonstrated that language models can produce verbal confidence estimates that reasonably correspond to prediction accuracy without requiring access to internal probability distributions or logits. These findings established verbally elicited confidence as a practical method for evaluating calibration, particularly for proprietary black-box models where internal

probabilities are unavailable.

Alternative approaches estimate uncertainty through multiple model generations rather than explicit self-report. Kuhn et al. (2023) introduced semantic entropy, arguing that uncertainty should be measured over underlying semantic meanings rather than individual token probabilities because multiple textual responses may express the same answer. Farquhar et al. (2024) further demonstrated that semantic entropy provides an effective signal for detecting hallucinations and confabulations in language models. Although these methods provide informative measures of uncertainty, they generally require repeated sampling and additional inference procedures that differ from the single confidence judgments presented to users during typical interactions.

Metacognition and Self-Knowledge in Language Models

Beyond uncertainty estimation, recent work has examined whether language models possess metacognitive awareness of their own knowledge limitations. Yin et al. (2023) introduced the SelfAware benchmark to evaluate whether LLMs can recognize questions that are fundamentally unanswerable or unknowable. Although prompting and instruction tuning improved performance, language models remained substantially less reliable than human participants in recognizing the limits of their knowledge. These findings suggest that language models exhibit partial self-awareness but continue to express unwarranted confidence under conditions of uncertainty. However, recognizing when a question cannot be answered differs from accurately calibrating confidence across questions that remain answerable but vary systematically in difficulty.

Human Trust in Language Model Confidence

The effectiveness of confidence estimates ultimately depends on how users interpret them. Steyvers et al. (2025) demonstrated that people frequently overestimate the accuracy of language model responses and that longer or more detailed explanations can increase perceived reliability without improving actual correctness. Their findings indicate that

miscalibrated confidence influences not only model evaluation but also user decision making by affecting when individuals choose to trust or defer to model outputs. Consequently, improving the calibration of communicated confidence has important implications for both model reliability and human–AI interaction.

Research Gap

Collectively, this body of work demonstrates that language model confidence is frequently imperfect, particularly under more challenging evaluation conditions (Desai & Durrett, 2020; Jiang et al., 2021). It also shows that verbally reported confidence provides a practical alternative to probability-based calibration for black-box models (Lin et al., 2022), while semantic uncertainty methods offer powerful sampling-based approaches for uncertainty estimation (Kuhn et al., 2023; Farquhar et al., 2024). Additional research has examined metacognitive self-awareness (Yin et al., 2023) and the influence of confidence communication on human trust (Steyvers et al., 2025). However, comparatively little attention has been given to how verbally reported confidence changes as task difficulty systematically increases across a standardized knowledge benchmark. In particular, few studies compare proprietary and open-weight language models under a common evaluation protocol that elicits explicit confidence judgments while controlling for question difficulty.

Purpose of the Present Study

The present study addresses this gap by evaluating verbal confidence calibration across pre-registered levels of task difficulty using a frozen, difficulty-stratified subset of the Massive Multitask Language Understanding (MMLU) benchmark. Claude Haiku, Llama 3.1, and Mistral are evaluated under a common answer-plus-confidence prompting protocol, and calibration is quantified using accuracy, the overconfidence gap, and Expected Calibration Error (ECE). By examining confidence as a function of task difficulty rather than overall benchmark performance alone, this study evaluates whether verbally reported confidence remains a reliable indicator of

correctness as questions become increasingly challenging and whether the resulting patterns resemble the difficulty-dependent confidence effects documented in human judgment.

3. METHODOLOGY

Experimental Design

This study evaluated the calibration of verbally reported confidence in large language models using a controlled, difficulty-stratified evaluation protocol. Three language models—Claude Haiku, Llama 3.1, and Mistral—were presented with the same frozen subset of multiple-choice questions from the Massive Multitask Language Understanding (MMLU) benchmark and required to provide both an answer and an explicit confidence estimate. Calibration was evaluated both overall and across pre-registered levels of task difficulty to determine whether confidence became increasingly misaligned with correctness as question difficulty increased.

Question Bank

Evaluation questions were drawn from the test split of the Massive Multitask Language Understanding (MMLU) benchmark (Hendrycks et al., 2021) using the Hugging Face `cais/mmlu` dataset. To ensure broad disciplinary coverage while maintaining a balanced experimental design, questions were restricted to three academic domains: STEM, Social Sciences, and Humanities. Subject categories were grouped into these domains using a predefined configuration file (`config/domains.yaml`).

Each subject was assigned to one of three difficulty levels (easy, medium, or hard) using a pre-registered difficulty mapping (`config/difficulty_map.yaml`) based on published MMLU subject-level performance patterns rather than the performance of the evaluated models. Difficulty assignments remained fixed throughout the study.

Within each domain–difficulty combination, up to 20 questions were randomly sampled using a fixed random seed (42), producing a frozen evaluation set of 180 questions (3 domains $\times$ 3 difficulty levels $\times$ 20 questions). The question bank was generated before model evaluation

began and was not modified thereafter, ensuring that every model received identical inputs under identical conditions.

Evaluated Models

Three widely used language models representing both proprietary and open-weight systems were evaluated under matched decoding conditions. Table 1 summarizes the evaluated models.

Table 1. Language models evaluated in this study.

Model	Access Method	Identifier
Claude Haiku	Anthropic API	claude-haiku-4-5-20251001
Llama 3.1	Ollama (local)	llama3.1
Mistral	Ollama (local)	mistral

All generations used a temperature of 0.3 with a maximum output length of 256 tokens. Each model completed two independent evaluation trials for every question, resulting in a planned total of 1,080 responses (180 questions $\times$ 3 models $\times$ 2 trials). All source code used for data collection, parsing, and analysis is publicly available at GitHub repository.

Confidence Elicitation

To ensure comparable confidence estimates across models, every inference call used the same structured prompt. Each model was instructed to return only the following format:

ANSWER: [A/B/C/D]
CONFIDENCE: [0–100]

The prompt contained the question, four labeled answer choices, explicit formatting instructions, and a valid response example. This constrained prompting protocol minimized formatting variability across models while avoiding chain-of-thought reasoning as a potential confounding factor. Confidence was therefore measured through a standardized verbal self-report rather than internal probability estimates.

Response Parsing

Model outputs were automatically parsed to extract both the selected answer and an integer confidence value between 0 and 100. Responses entered the calibration analysis only when both fields were successfully parsed (`parse_ok = True`). Unparseable responses were retained in

the raw dataset but excluded from all calibration analyses.

Final parse rates are summarized in Table 2.

Table 2. Response parse rates by model.

Model	Parseable Responses
Claude Haiku	267 / 360 (74.2%)
Llama 3.1	360 / 360 (100.0%)
Mistral	359 / 360 (99.7%)

Most Claude Haiku parsing failures occurred when verbose reasoning exceeded the maximum output length before the required response tags were produced. Mistral produced one free-form response rather than the required multiple-choice format.

Evaluation Metrics

Correctness was determined by comparison with the official MMLU answer key. Confidence values were normalized to the interval $[0, 1]$ by dividing reported percentages by 100.

Calibration was evaluated using four primary metrics:

- **Accuracy**, defined as the proportion of correctly answered questions.
- **Mean confidence**, computed as the average normalized confidence.
- **Overconfidence gap**, defined as mean confidence minus accuracy, where positive values indicate overconfidence and negative values indicate underconfidence.
- **Expected Calibration Error (ECE)**, computed using ten equally spaced confidence bins. An ECE of 0 indicates perfect calibration (e.g., predictions made with 80% confidence are correct approximately 80% of the time), whereas larger values indicate increasing mismatch between reported confidence and observed accuracy.

To quantify the relationship between task difficulty and calibration, ordinary least squares regression was performed using cell-level overconfidence gaps as the dependent variable and difficulty coded numerically (easy = 1,

medium = 2, hard = 3) as the independent variable. Domain-specific summaries for STEM, Social Sciences, and Humanities were also computed as secondary analyses.

Statistical Analysis

The primary analysis examined whether overconfidence increased systematically across the pre-registered difficulty levels. Calibration metrics were computed for each model overall and separately within each difficulty tier. Human findings regarding the *hard–easy effect* served only as a qualitative comparison and were not interpreted as evidence that language models share the same underlying cognitive mechanisms.

Results are presented using reliability diagrams, overconfidence-by-difficulty plots, and Expected Calibration Error comparisons across models and academic domains.

Reproducibility

To facilitate replication, all experimental configurations, the frozen evaluation set, raw model outputs, processed datasets, and figure-generation scripts are included in the project repository. Because the evaluation bank and sampling procedure are fixed, future studies can reproduce the analysis directly or extend it to additional language models while maintaining an identical evaluation protocol.

4. RESULTS

All analyses reported in this section include only successfully parsed responses containing both a valid multiple-choice answer and an integer confidence score between 0 and 100. Across the planned 1,080 model responses, 986 met these criteria and were included in the calibration analyses. Overall performance metrics are presented in Table 3, while calibration trends across difficulty and domain are presented in subsequent sections.

Overall Performance

Table 3. Overall performance metrics by model.

Model	N	Accuracy	Mean Conf.	Gap	ECE
Claude Haiku	267	0.921	0.901	−0.020	0.048
Llama 3.1	360	0.636	0.867	0.231	0.231
Mistral	359	0.538	0.979	0.441	0.446

Table 3 summarizes accuracy, mean confidence, overconfidence gap, and Expected Calibration Error (ECE) for each evaluated model. Claude Haiku achieved the highest accuracy while maintaining the lowest calibration error, resulting in a slight aggregate underconfidence. In contrast, both Llama 3.1 and Mistral reported confidence levels that exceeded their observed accuracy, with Mistral exhibiting the largest confidence–accuracy mismatch.

Calibration Across Task Difficulty

Table 4. Performance metrics by model and task difficulty.

Model	Difficulty	N	Accuracy	Mean Conf.	Gap	ECE
Claude Haiku	Easy	99	0.939	0.912	−0.027	0.052
Claude Haiku	Medium	98	0.929	0.876	−0.053	0.087
Claude Haiku	Hard	70	0.886	0.922	0.037	0.072
Llama 3.1	Easy	120	0.758	0.875	0.117	0.117
Llama 3.1	Medium	120	0.642	0.858	0.216	0.216
Llama 3.1	Hard	120	0.508	0.868	0.360	0.360
Mistral	Easy	120	0.583	0.978	0.395	0.395
Mistral	Medium	120	0.625	0.970	0.345	0.360
Mistral	Hard	119	0.403	0.988	0.585	0.585

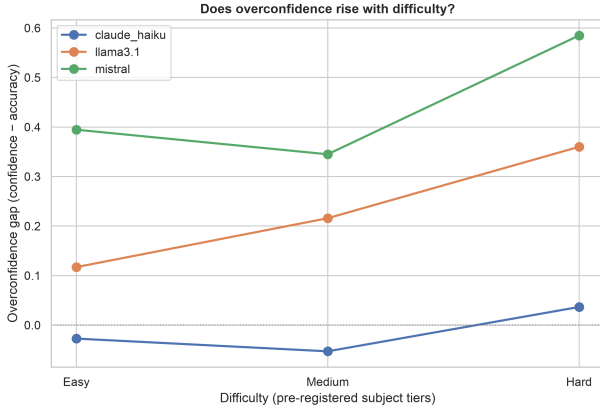


Figure 1. Overconfidence gap across task difficulty levels.

Table 4 and Figure 1 show calibration performance across easy, medium, and hard questions. For all three models, confidence became progressively less aligned with empirical accuracy as task difficulty increased. Llama 3.1 displayed a monotonic increase in overconfidence across difficulty levels, while Mistral exhibited the largest calibration errors overall. Claude Haiku remained comparatively well calibrated, transitioning from slight underconfidence on easier questions to mild overconfidence on harder questions.

Domain-Level Performance

Calibration performance also varied across

Table 5. Expected calibration error by model and domain.

Model	Domain	ECE	N
Claude Haiku	Humanities	0.065	97
Claude Haiku	STEM	0.075	81
Claude Haiku	Social Science	0.072	89
Llama 3.1	Humanities	0.128	120
Llama 3.1	STEM	0.413	120
Llama 3.1	Social Science	0.152	120
Mistral	Humanities	0.291	120
Mistral	STEM	0.656	119
Mistral	Social Science	0.392	120

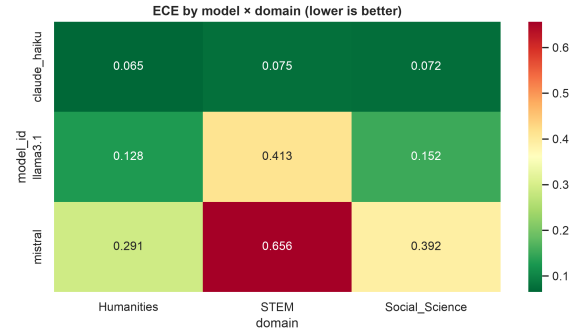


Figure 2. ECE heatmap by model and domain.

academic domains (Table 5 and Figure 2). Claude Haiku maintained consistently low calibration error across STEM, Social Sciences, and Humanities. In contrast, both open-weight models exhibited their greatest miscalibration on STEM questions and their lowest calibration error on Humanities questions, with Social Sciences producing intermediate results.

Reliability Diagrams

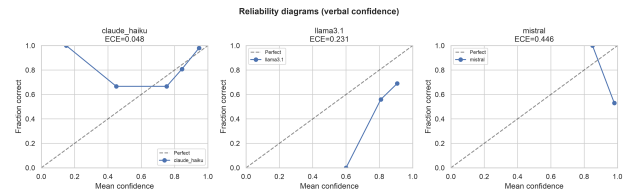


Figure 3. Reliability diagrams comparing model confidence calibration.

Figure 3 illustrates the relationship between reported confidence and observed accuracy. Claude Haiku produced reliability curves closest to the ideal calibration diagonal, indicating strong agreement between confidence and empirical performance. Llama 3.1 showed increasing

deviation at higher confidence levels, while Mistral demonstrated the largest mismatch between reported confidence and observed accuracy.

Relationship Between Difficulty and Overconfidence

Table 6. Regression of overconfidence gap on task difficulty.

Model	Slope	R^2	p-value	Hard–Easy Corr.
Claude Haiku	0.032	0.479	0.513	0.692
Llama 3.1	0.121	0.988	0.068	0.994
Mistral	0.095	0.565	0.459	0.751

Regression results are presented in Table 6. All three models produced positive slopes relating task difficulty to overconfidence, indicating that confidence became less aligned with empirical accuracy as questions became more difficult. Llama 3.1 exhibited the strongest relationship between difficulty and overconfidence, while Claude Haiku demonstrated the weakest trend due to its relatively stable calibration across difficulty levels.

5. DISCUSSION

The primary empirical finding is that all three evaluated models exhibit increased overconfidence as task difficulty increases (Table 4; Figure 1). Although accuracy consistently declines on harder questions, reported confidence does not decrease proportionally, producing larger confidence–accuracy discrepancies precisely in settings where model performance is weakest. This pattern indicates that confidence estimates are not fully sensitive to increasing task difficulty. Claude Haiku represents a notable exception in magnitude: while it follows the same directional trend, its calibration error remains comparatively small, shifting from slight underconfidence on easy items (-0.027) to mild overconfidence on hard items ($+0.037$).

Substantial differences emerge across models. Claude Haiku achieves the strongest combination of predictive accuracy and calibration, obtaining an overall accuracy of 0.921, an Expected Calibration Error (ECE) of 0.048, and an aggregate confidence gap of only -0.020 (Table 3). In contrast, Llama 3.1 demonstrates

moderate accuracy (0.636) while exhibiting meaningful overconfidence (gap = 0.231, ECE = 0.231). Mistral displays the most severe calibration failure: despite achieving the lowest accuracy among evaluated models (0.538), it maintains near-ceiling confidence (0.979), resulting in the largest confidence–accuracy gap (0.441) and highest ECE (0.446). These differences are also reflected in the reliability diagrams (Figure 3), where Claude Haiku remains closest to the ideal calibration diagonal, whereas Llama 3.1 and Mistral increasingly deviate from observed accuracy at higher confidence levels.

The analysis of domain-level calibration reveals that these failures are not uniformly distributed across subject areas. Both open-weight models demonstrate their largest calibration errors on STEM questions, with ECE values of 0.413 for Llama 3.1 and 0.656 for Mistral (Table 5; Figure 2). Conversely, Claude Haiku maintains consistently low calibration error across Humanities, Social Science, and STEM domains (ECE range: 0.065–0.075). These results suggest that calibration quality depends not only on overall model capability but also on the interaction between model characteristics and the type of reasoning required by the task.

The observed relationship between difficulty and overconfidence is directionally consistent with findings in human judgment research, where individuals often exhibit greater confidence errors when evaluating more challenging decisions (Moore and Healy, 2008). However, these results should not be interpreted as evidence that language models exhibit the same cognitive mechanisms underlying human overconfidence phenomena such as the Dunning–Kruger effect (Kruger and Dunning, 1999). Instead, the empirical parallel is narrower: as evaluation difficulty increases, model confidence becomes less aligned with actual correctness. This distinction is important because language model confidence reflects generated self-assessments rather than human metacognitive awareness.

Compared with previous studies of language model confidence estimation, these findings occupy an intermediate position between optimistic assessments of model self-evaluation

and reports of systematic overconfidence. Prior work has demonstrated that stronger models can produce useful confidence signals under appropriate elicitation methods (Kadavath et al., 2022; Lin et al., 2022; Tian et al., 2023). At the same time, other studies have identified substantial miscalibration in verbalized confidence, particularly among weaker models and challenging tasks (Desai and Durrett, 2020; Xiong et al., 2024). The contribution of this study is the explicit examination of how calibration changes with task difficulty and domain. The results demonstrate that open-weight models experience rapidly increasing confidence errors on difficult questions, whereas Claude Haiku remains comparatively stable across difficulty levels (Figure 1).

These findings have practical implications for users interacting with language models. A reported confidence score is attractive because it provides a simple and interpretable indicator of uncertainty. However, our results demonstrate that the reliability of this signal varies substantially by model. For Claude Haiku, verbal confidence provides a reasonably informative approximation of correctness, whereas for Mistral, high confidence frequently corresponds to incorrect predictions. Consequently, users should interpret high-confidence responses more cautiously when models demonstrate weaker calibration, particularly on difficult items and STEM-focused tasks. Confidence estimates should therefore complement, rather than replace, external verification mechanisms.

For trustworthy AI deployment, calibration evaluation should extend beyond aggregate metrics. A model may achieve acceptable overall calibration while remaining severely overconfident in specific conditions where users are most likely to rely on its outputs. Evaluations should therefore consider calibration across dimensions such as difficulty, domain, and model capability rather than relying exclusively on global ECE values. Approaches that improve uncertainty estimation (Lin et al., 2022; Kuhn et al., 2023; Farquhar et al., 2024) and interfaces that communicate uncertainty more effectively to users (Steyvers et al., 2025) represent complementary directions

toward safer human–AI interaction. Overall, verbal confidence remains a useful and low-cost interface signal, but it is inherently imperfect and should be interpreted alongside model reliability, task characteristics, and independent validation.

6. LIMITATIONS

This study has several limitations that define the scope of its conclusions. First, evaluation is restricted to multiple-choice questions from the MMLU benchmark. Although this setting enables controlled measurement of accuracy and confidence calibration, the findings may not generalize directly to open-ended generation, tool-augmented agents, or interactive conversational systems where uncertainty expression may operate differently. Additionally, question difficulty is assigned using a pre-registered subject-level mapping rather than estimated at the individual item level through approaches such as item response theory (IRT). Consequently, variation in difficulty within each subject category is compressed into three ordinal levels, potentially obscuring finer-grained relationships between task difficulty and calibration.

A second limitation concerns the measurement of confidence. This study evaluates verbal confidence scores, represented as integer values from 0 to 100, rather than model-internal probability estimates or token-level likelihoods. Verbalized confidence provides an accessible approximation of uncertainty for human-facing applications, but it may not correspond directly to the underlying predictive distribution of a language model. Prior work has demonstrated that confidence estimates can vary substantially depending on elicitation strategy, model architecture, and evaluation methodology. Therefore, these results should be interpreted as an analysis of externally expressed confidence rather than a direct measurement of internal uncertainty.

The number of evaluated models and trials also limits the breadth of the conclusions. This study examines three language models with two evaluation trials per question. Expanding the analysis to additional model families, larger

benchmark sets, and more repeated trials would provide stronger estimates of calibration stability and reduce uncertainty associated with individual model behaviors. Future work should also examine whether the observed difficulty-dependent calibration patterns persist across additional architectures and model scales.

A further consideration is response format compliance. Claude Haiku produced 267 usable responses out of 360 total attempts (74.2% parse rate), primarily because some responses exceeded the required output format and failed to provide the expected ANSWER and CONFIDENCE fields within the imposed token limit. The calibration analyses therefore condition on successfully parsed responses rather than correcting missing outputs through imputation. This approach preserves the integrity of the evaluated confidence signals but may introduce selection effects if format compliance is systematically related to model uncertainty or task difficulty. Llama 3.1 achieved complete parse compliance, while Mistral contributed 359 usable responses due to a single invalid answer format.

Finally, the interpretation of these findings should remain appropriately bounded. Although the observed trends are directionally consistent with human research on difficulty-related overconfidence, this study does not claim that language models possess human psychological mechanisms or metacognitive processes. The reported results are specific to the evaluation configuration used here, including the temperature setting of 0.3, the fixed 180-item question set, the constrained prompting strategy, and the July 2026 model versions evaluated. Future studies should examine whether these calibration patterns remain consistent under alternative prompting strategies, sampling conditions, and model updates.

7. FUTURE WORK

The findings of this study motivate several directions for future investigation. First, future evaluations should extend beyond the current MMLU-based setting to more challenging and complementary benchmarks, such as

MMLU-Pro, GPQA, and ARC. These datasets would help determine whether the observed relationship between increasing difficulty and overconfidence persists in settings where baseline accuracy is substantially lower and reasoning demands are greater.

Second, broader model coverage is necessary to evaluate whether the calibration patterns observed here generalize across additional frontier and open-weight language models. In particular, examining a wider range of model capabilities and architectures would clarify whether near-calibration is consistently associated with stronger models or whether calibration behavior depends on model-specific training and alignment procedures.

Third, future work should compare verbal confidence estimates with model-internal uncertainty measures, such as token-level probabilities or log probabilities, on identical evaluation items where such signals are available. This comparison would help determine whether expressed confidence reflects underlying uncertainty or represents a separate communication behavior influenced by prompting and interface design.

Fourth, the effect of confidence elicitation strategies should be investigated systematically. Alternative prompting approaches, including chain-of-thought-enabled reasoning, top- k confidence estimates, and multi-stage confidence reporting, could reveal how different interaction formats influence calibration while holding the evaluation set constant. Similarly, calibration improvement techniques such as temperature scaling, consistency-based aggregation, and semantic-entropy-based uncertainty estimation should be evaluated across the same difficulty and domain partitions used in this study.

Fifth, future studies should examine the relationship between model calibration and human reliance behavior. Human-subject experiments measuring trust, verification behavior, and error detection could connect quantitative calibration metrics such as ECE and overconfidence gap with real-world user decision-making. Such studies would provide insight into when confidence estimates improve

human–AI interaction and when they introduce additional risk.

Finally, future evaluations should incorporate finer-grained measures of task difficulty and a broader range of calibration metrics. Replacing subject-level difficulty categories with item-level estimates derived from item response theory or empirical human performance would provide more precise analyses of difficulty-dependent calibration. Additionally, metrics beyond ECE and confidence–accuracy gap, including Brier score, selective prediction risk, and semantic entropy, could provide a more complete characterization of uncertainty quality in language models.

8. CONCLUSION

Language models are often trusted because their responses are fluent and persuasive; however, whether their stated confidence provides a reliable signal of correctness is an empirical question. This study evaluated the calibration behavior of Claude Haiku, Llama 3.1, and Mistral using a frozen, difficulty-stratified subset of MMLU consisting of 180 questions across three domains and three difficulty levels. Under a controlled verbal confidence elicitation protocol, models were evaluated using accuracy, confidence–accuracy gap, Expected Calibration Error (ECE), and difficulty-dependent calibration trends.

Among successfully parsed responses, Claude Haiku demonstrated the strongest combination of accuracy and calibration, achieving an accuracy of 0.921, an ECE of 0.048, and an overall confidence gap of -0.020 . In contrast, Llama 3.1 and Mistral exhibited substantially larger calibration errors, with confidence gaps of 0.231 and 0.441 and ECE values of 0.231 and 0.446, respectively. Across all evaluated models, overconfidence increased as task difficulty increased, with the largest degradation observed among the open-weight models (Figure 1; Table 4). Domain-level analysis further showed that calibration failures were concentrated in STEM tasks for Llama 3.1 and Mistral, while Claude Haiku maintained consistently low ECE across domains (Figure 2;

Table 5).

These results demonstrate that verbal confidence is not a universally reliable indicator of model correctness. Its usefulness depends strongly on model capability, task difficulty, and domain. For stronger models, confidence estimates may provide a meaningful approximation of uncertainty, while weaker models may produce highly confident responses despite substantial error rates. Therefore, confidence scores should be treated as supporting signals rather than substitutes for verification, retrieval, or human review.

The central conclusion of this work is that calibration must be evaluated where models are most likely to fail, not only where average performance appears strong. Under a fixed verbal confidence protocol, increasing task difficulty consistently amplifies overconfidence, and the severity of this failure varies substantially across models. Confidence without corresponding competence represents a measurable reliability risk for LLM deployment, emphasizing the need for calibration-aware evaluation and uncertainty estimation methods.

Figures and Tables

The following figures and tables summarize the primary experimental results:

- **Figure 3:** Reliability diagrams comparing predicted confidence with observed accuracy across evaluated models.
- **Figure 1:** Overconfidence trends across task difficulty levels, illustrating how confidence–accuracy gaps change from easy to hard questions.
- **Figure 2:** Domain-level calibration heatmap showing Expected Calibration Error (ECE) across Humanities, STEM, and Social Science categories.
- **Table 3:** Overall model performance, including accuracy, mean confidence, confidence–accuracy gap, and Expected Calibration Error.
- **Table 4:** Calibration performance across easy, medium, and hard question categories.

- **Table 5:** Expected Calibration Error values across academic domains for each model.
- **Table 6:** Regression analysis measuring the relationship between task difficulty and overconfidence.

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